F79SU Survival Models 2025-2026 Coursework 1

Comparing lifetimes of feline breeds for pet insurance analysis

Scenario: Your company wishes to establish a line of pet insurance products and thus require some preliminary analysis of the lifetimes of two particular feline breeds: the American Shorthair and the British Shorthair. The data collection department has provided you with a set of data consisting of lifetimes of such cats. Your task is to answer questions posed to you by various colleagues and supervisors using this dataset.

Dataset: The R-script `Assignment1.R` (available on Canvas) generates the lifetimes (some censored) for American Shorthairs and British Shorthairs, as well as the age at contraction of a terminal disease called Feline Infectious Peritonitis (FIP). The lifetimes are stored in `Lifetime` (in years), censoring information in `Censor` (0 indicates that the observation was censored, 1 indicates that the observation was *not* censored), breed type in `Breed` (1 indicates American Shorthair, 2 indicates British Shorthair) and age at contraction of FIP in `FIPAge` (in years old). Note that a value of 0 in `FIPAge` indicates that the corresponding feline never contracted FIP.

Data generation: You should set the argument to `set.seed` as the last 4 digits of your Heriot-Watt student ID number. For example, if your student ID is H12345678 then your seed would be 5678. Then, generate your data by running the script. Aside from the number for the seed, do not change any of the code above the comment line reading 'Please insert your R code after this line'. **Failure to follow this data generation process exactly may result in 0 marks being awarded.**

Multiple-Choice Questions: In some of the following tasks, you may be asked to answer multiple-choice questions with options (A), (B), (C) and (D) — among these, choose the most accurate option. In each of these questions, please do the following in your R Script:

If your answer is (A), set the relevant quantity to 1 (e.g. `Quantity1 <- 1`).
If your answer is (B), set the relevant quantity to 2 (e.g. `Quantity1 <- 2`).
If your answer is (C), set the relevant quantity to 3 (e.g. `Quantity1 <- 3`).
If your answer is (D), set the relevant quantity to 4 (e.g. `Quantity1 <- 4`).

Failure to do so may result in 0 marks being awarded.

R Script: Your R Script should demonstrate your workings sufficiently clearly where calculations or user-defined functions are required.

Excel Tabulation: You should clearly tabulate your answers in a new Excel workbook. Populate the cells A1 to A22 with the text strings `Quantity1`, `Quantity2`, `Quantity3`, ..., `Quantity22`. Also, populate the cells B1 to B22 with your corresponding answers (in numerical values, to reasonable precision) for each quantity. For non-numerical quantities such as the functions required in `Quantity11` and `Quantity18`, you may leave the corresponding cell blank. **Failure to do so may result in 0 marks awarded.** Your numerical answers in the Excel workbook should reasonably match the outputs of your R Script; otherwise you may also be penalised.

Part 1

As a first step of analysis of the lifetimes of the cats in your dataset, you are asked to construct Kaplan-Meier estimates of the survivor functions for American Shorthairs, for British Shorthairs, and for all cats from both breeds involved in the study. You should do this using the `survfit` function from the `survival` library in R.

In considering a pet life insurance contract where a claim is made upon the associated pet's death, your coworkers in the pricing department would like you to:

1. Estimate the probability that an American Shorthair survives for at least 10 years.
   Store this as `Quantity1` in your R Script and Excel file (0.5 marks).

2. Provide an equal-tailed 95% confidence interval for your estimate in Question 1.
   Store the lower limit as `Quantity2` and the upper limit as `Quantity3` (0.5 marks).

3. Estimate the probability that a British Shorthair dies within 12 years.
   Store this as `Quantity4` (0.5 marks).

4. Provide an equal-tailed 95% confidence interval for your estimate in Question 3.
   Store the lower limit as `Quantity5` and the upper limit as `Quantity6` (0.5 marks).

Note that `NA` is a possible answer where appropriate.

Prospective clients may take up such insurance policies midway through their pets' lives, hence estimates related to *remaining* lifetimes are also important. You are asked to:

5. Estimate the probability that a client with an American Shorthair makes a claim within 5 years, given that the cat is currently 3 years old.
   Store this as `Quantity7` (0.5 marks).

6. Estimate the probability that a client with a British Shorthair makes a claim within 5 years, given that the cat is currently 3 years old.
   Store this as `Quantity8` (0.5 marks).

You receive a new data point from the data collection department. This cat passed away within 8.2 years of age but you notice that the `Breed` entry is empty. No one is currently available to provide you with this information and you have to fill in this missing entry yourself.

7. Based on the Kaplan-Meier estimates constructed above, what is the probability that this cat was an American Shorthair, assuming the data you have indicates relative population proportions?
   Store your answer as `Quantity9` (1 mark).

Note that you do *not* need to append this new data point to the dataset generated.

[Part 1 Total: 4 marks]

Part 2

Subsequently, your supervisor asks you to consider a Cox proportional hazards model, using American Shorthairs' lifetimes as the baseline. The hazard function $\lambda_B(x)$ for a British Shorthair aged x years can then be written as

$$\lambda_B(x) = \lambda_A(x)\exp(\beta)$$

where $\lambda_A(x)$ is the hazard function for an American shorthair aged x years, and $\beta \in \mathbb{R}$.

1. You come across a previous study that reported a significant estimate for β to be -0.25. What does this suggest about the lifetimes of American and British Shorthairs?

   (A) American Shorthairs live longer than British Shorthairs.
   (B) British Shorthairs live longer than American Shorthairs.
   (C) American Shorthairs are more likely to live past any age than British Shorthairs.
   (D) British Shorthairs are more likely to live past any age than American Shorthairs.

   Store your answer in `Quantity10` (1 mark).

To further analyze the dataset, your coworkers require the likelihood profile of, and statistical tests relating to, β under the Cox proportional hazards model for the dataset given above. Therefore, you are asked to:

2. Code an R function for the partial log-likelihood of β under the Cox proportional hazards model without removing any constant terms. Hence, evaluate this partial log-likelihood at $\beta = 0.1$.
   Name this function as `Quantity11` and store the value of the partial log-likelihood at $\beta = 0.1$ as `Quantity12` (2 marks).

3. Using this function or otherwise, obtain $\hat{\beta}$, the maximum likelihood estimate of β.
   Store this as `Quantity13` (1 mark).

4. Conduct the likelihood ratio test (LRT) on β.
   Store the test statistic as `Quantity14` and the p-value as `Quantity15` (1 mark).

5. Based on the LRT in Question 4, at a 2% significance level, which of the following insights about the lifetimes of American Shorthairs and British Shorthairs is the most accurate?

   (A) β is significantly different from 0, suggesting a significant difference between the lifetimes of American and British Shorthairs.
   (B) β is not significantly different from 0, suggesting no significant difference between the lifetimes of American and British Shorthairs.
   (C) β is significantly different from $\hat{\beta}$, suggesting a significant difference between the lifetimes of American and British Shorthairs.
   (D) β is not significantly different from $\hat{\beta}$, suggesting no significant difference between the lifetimes of American and British Shorthairs.

   Store your answer in `Quantity16` (1 mark).

Upon further research, you identify that a terminal disease called Feline Infectious Peritonitis (FIP) significantly impacts the lifetime of both feline breeds.

6. Without looking into specific ages at FIP contraction in the dataset, is the Cox proportional hazards model detailed above still a suitable model for your analysis?

   (A) Yes, we may simply add a covariate related to the presence of FIP to the Cox proportional hazards model via $\lambda_B(t) = \lambda_A(t)\exp(\beta_1 z_1 + \beta_2 z_2)$, where $\beta_1, \beta_2 \in \mathbb{R}$ and

$$z_1 = \begin{cases} 0 & \text{if breed is American Shorthair} \\ 1 & \text{if breed is British Shorthair} \end{cases} \qquad \text{and}$$

$$z_2 = \begin{cases} 0 & \text{if cat contracts FIP during its lifetime} \\ 1 & \text{if cat does not contract FIP during its lifetime.} \end{cases}$$

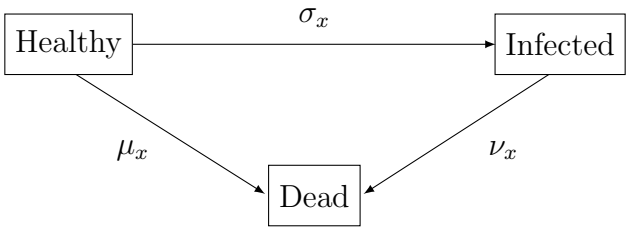
   (B) Yes, as long as both breeds' instantaneous rates of contracting FIP stays approximately the same over all ages.

   (C) No, since the ratio of hazards $\lambda_B(x)/\lambda_A(x)$ will no longer be a constant expressible by $\exp(\beta)$, as specified in the Cox proportional hazards model.

   Store your answer in `Quantity17` (1 mark).

[Part 2 Total: 7 marks]

Part 3



To account for the variability in cats' lifetimes due to FIP, you decide to examine a 3-state Markov model shown in the diagram above. In the Healthy (H) state, the cat associated with the insurance policy is alive and well; in the Infected (I) state, the cat is alive but infected with FIP; in the Dead (D) state, the cat is dead, upon which a claim is made by its owner. At exact age x, a cat's transition intensity from states H to I, H to D, and I to D are σ_x, μ_x, and ν_x, respectively. This model is applicable to all cat breeds, though the specific forms of σ_x, μ_x and ν_x may be different across breeds.

Note that the ages at which each cat was first infected with FIP is given in the dataset, in the R variable named `FIPAge`; you are reminded that a value of 0 in `FIPAge` for a feline indicates no contraction of FIP throughout its period of observation. You wish to use this dataset to infer ν_x. For simplicity, you first assume constant transition intensities $\mu_x = \mu, \sigma_x = \sigma$ and $\nu_x = \nu$ for all feline ages $x \geq 0$.

1. Code an R function for the log-likelihood function of (μ, σ, ν) without removing any constant terms. Hence, evaluate this log-likelihood at $(\mu, \sigma, \nu) = (0.1, 0.1, 0.1)$.
   Store this function as `Quantity18` and the value of the log-likelihood at $(\mu, \sigma, \nu) = (0.1, 0.1, 0.1)$ as `Quantity19` (1 mark).

2. Using this function or otherwise, calculate $\hat{\nu}$, the maximum likelihood estimate for ν.
   Store this as `Quantity20` (1 mark).

For subsequent tasks, to retain the flexibility of the Markov model, assume non-constant transition intensities μ_x, σ_x and ν_x.

Coincidentally, you came across an adoption post for a 7 year-old Siamese cat. If you were to adopt this cat, you plan to buy an insurance policy from your company. In deciding the worthiness of such a policy, you would naturally want to consider all possibilities including FIP contraction. A research article suggests that the random variable representing the age at FIP contraction for Siamese cats has a survival function of the form:

$$S(x) = \exp\left(- ax - \frac{1}{2}bx^2 \right),$$

for $x \geq 0$, where $a, b \in \mathbb{R}$. Under this model:

3. Which form of σ_x is the most suitable for the Markov model defined above?
   (A) $\sigma_x = a \log x + b$
   (B) $\sigma_x = a + bx$
   (C) $\sigma_x = \exp(a + bx)$
   (D) $\sigma_x = ax + \frac{1}{2}bx^2$
   Store your answer in `Quantity21` (1 mark).

4. Calculate the probability that this new adopted Siamese cat contracts FIP within the next 3 years for $a = 0.1$ and $b = 0.08$.
   Store this as `Quantity22` (1 mark).

[Part 3 Total: 4 marks]

Further Points to Note

Questions for the Lecturers: If you have any questions on the assignment, please ask in the discussion board rather than e-mailing lecturers directly - that way, all students will see the same response.

Collaboration and Plagiarism: Students are allowed to discuss the methods used with other students, but your submitted project must be **all your own work**. Particularly, students must never give hard or soft copies of their coursework reports or code to another student, and must always refuse any such requests; otherwise, if such collusion is detected, this will be reported to the School's Discipline Committee. If found guilty of collusion, the penalty could involve voiding the course.

Generative Aritifical Intelligence: Generative Artificial Intelligence (GenAI) is not permitted for any part of this assignment. Unauthorised usage is considered a form of collusion.

R Script: There are 15 marks in total. Your R Script must contain all the relevant quantities, with variables defined as instructed, and you must ensure that your code runs without generating errors. **Failure to do so may result in 0 marks being awarded**. You can check that your code runs correctly by running it through the file `TestSubmissionAssignment1.R`. In your R Script, please do not give references to directories on your own computer (e.g. specifying a directory with `setwd`) or make use of any R library aside from the `survival` library.

Answer Tabulation in Excel: You must clearly tabulate your answers in an Excel file, with one column showing the relevant quantities (e.g. `Quantity13`) and another column showing the corresponding answers in numerical value, as instructed in the first page of this assignment sheet.

Submission: You must **submit your R Script and Excel file** using the links on Canvas, and **not** by email or handed in as a paper copy. You should also complete the Student Declaration of Authorship at the same time and submit to the appropriate link.

Deadline: The deadline for submission is 9:00 am UK time = 1:00 pm Dubai time = 5:00 pm Malaysia time on Friday 13th February 2026 (Week 5); projects may be submitted early. Late projects will be marked in line with the University Coursework Policy, i.e. a standard deduction of 30% if submitted within 5 working days and 0 marks awarded if submitted later. You can, however, submit a mitigating circumstances application, but these may only be applied at the end of the semester. No individual extensions are allowed.

Assessment: This coursework will contribute 15% to the final mark for the course.